

Automated Podcast Transcription and Topic Segmentation Using NLP

Overview

This project focuses on building an automated system that converts podcast or audio recordings into structured textual information. The system performs audio cleaning, speech-to-text transcription, text preprocessing, topic segmentation, keyword extraction, and visualization. Manual transcription and analysis of long audio files is time-consuming and inefficient. This project automates the entire pipeline, enabling users to quickly understand audio content through text and visual analytics. A user-friendly interface allows users to upload and select transcripts, listen to cleaned audio, view segmented text, search keywords, and analyze data using charts and word clouds.

Approach

- Audio Input – Raw audio files are taken from the dataset.
- Audio Cleaning – Noise removal and audio enhancement is performed to improve clarity.
- Speech Transcription – Cleaned audio is converted into text.
- Text Preprocessing – Removing unnecessary symbols, headers, and stop words.
- Topic Segmentation – Long transcripts are divided into meaningful segments.
- Keyword Extraction – Important words are extracted based on frequency.
- Visualization – Bar charts and word clouds for keyword distribution.
- User Interface – Streamlit-based interface for easy interaction.

Tech Stack

Programming Language: Python

UI Framework: Streamlit

Data Processing: Pandas

Visualization: Matplotlib, WordCloud

NLP: NLTK

Audio Processing: Librosa

IDE: VS Code

Version Control: GitHub

Setup

1. Install Python from <https://www.python.org> and add it to PATH.
2. Clone the repository using Git.
3. Install required libraries using pip.
4. Download NLTK data (punkt, stopwords).
5. Run the application using Streamlit.

Usage

- Launch the web application.
- Select a transcript file.
- View transcription text and cleaned audio.
- Analyze segmented text and keywords.
- View charts and word clouds.

Troubleshooting

pip not recognized – Use `python -m pip install`

Missing library error – Reinstall libraries

Streamlit not opening – Check Python PATH

Audio not loading – Verify file path

Visualization issues – Ensure matplotlib installed

Limitations

- Accuracy depends on audio quality.
- Large files increase processing time.
- Topic segmentation is rule-based.

- Limited language support.

Future Scope

- Real-time transcription
- Multi-language support
- Advanced AI topic modeling
- Automatic summarization
- Cloud deployment
- Speaker identification
- Sentiment analysis

References

Python Documentation
Streamlit Documentation
NLTK Library
Librosa Documentation
Matplotlib Documentation
WordCloud Library
GitHub Open Source Projects